THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS UNIT TEST SET A

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The supplement of 118° is: 1

(a) 28° **(b)** 242° **(c)** 118° **(d)** 62°

2. An angle greater than 90° and less than 180° is: 1

(a) right **(b)** acute **(c)** reflex **(d)** obtuse

3. In a linear pair, one angle is 112°. The other is: 1

(a) 248° **(b)** 22° **(c)** 68° **(d)** 112°

4. A letter puzzle: if 3A + 2 = 17 then A = 1

(a) 6 **(b)** 4 **(c)** 5 **(d)** 3

5. Kaprekar's constant for 4-digit numbers is: 1

(a) 6174 **(b)** 1729 **(c)** 1089 **(d)** 9999

6. In the Virahāṅka sequence 1, 2, 3, 5, 8, … the term after 13 is: 1

(a) 21 **(b)** 20 **(c)** 22 **(d)** 26

7. Sides 7 cm, 10 cm and 12 cm form a triangle? 1

(a) Cannot say **(b)** Yes **(c)** Only if equal **(d)** No

8. The perimeter of an equilateral triangle of side 7 cm is: 1

(a) 21 cm **(b)** 49 cm **(c)** 28 cm **(d)** 14 cm

9. In an isosceles triangle the vertex angle is 70°. Each base angle is: 1

(a) 110° **(b)** 70° **(c)** 20° **(d)** 55°

10. The area of a rectangle 2/3 m by 3/5 m is: 1

(a) 2/5 m² **(b)** 10/9 m² **(c)** 9/10 m² **(d)** 19/15 m²

11. 4/7 ÷ 2/3 = 1

(a) -2/21 **(b)** 7/6 **(c)** 6/7 **(d)** 8/21

12. 9/10 ÷ 3/5 = 1

(a) 3/2 **(b)** 3/10 **(c)** 27/50 **(d)** 2/3

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. If two intersecting lines make an angle of 70°, find the other three angles. 2

14. Is 2,35,716 even or odd? Why? 2

15. What is a palindrome? Give two examples. 2

16. Find the third angle of a triangle with angles 52° and 68°. 2

17. Is 4/5 ÷ 2/3 equal to 2/3 ÷ 4/5? Check. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Two co-interior angles are (2x + 15)° and (3x + 5)°. If lines are parallel, find x. 3

19. Find the sum of first 10 odd numbers and explain without adding. 3

20. A right triangle's acute angles differ by 20°. Find them. 3

21. Find the product 5/8 × 16/25 × 5/6. 3

22. A rope is 9 m long. How many pieces of 3/4 m can be cut and what remains? 3

OR

Divide 2/3 by 4 and 4 by 2/3. 3

23. State true or false with reason: Two acute angles can be supplementary. 3

OR

A ladder leans making 70° with the ground. Find the angle with the wall. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) State divisibility tests for 2, 3, 5, 9, 10.
(ii) Test 8,730.
(iii) Test 4,725. 5

OR

(i) Prove that the sum of two odd numbers is even.
(ii) Prove that odd × odd is odd.
(iii) Use these to decide the parity of 7 × 9 + 11. 5

25. **Case study:** A ribbon of 15 m is cut into pieces of 5/6 m for gift wrapping. 5
(i) Number of pieces.
(ii) If each piece costs ₹12, total.
(iii) Left if 4 pieces damaged.
(iv) Length wasted.

OR

(i) Find the reciprocal of 1 2/3. 5
(ii) Verify the product with the number is 1.
(iii) Find $(1\ 2/3)^2$.

END OF QUESTION PAPER — Total: 50 Marks